

Correlation Witnesses versus Magic: A k -Resolved Nonstabilizerness Map for $SU(2)_k$ Anyon Fusion Spaces

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Standard correlation witnesses — spatial Bell/CHSH inequalities, temporal Leggett–Garg K_3 , and KCBS contextuality — are known to detect nonstabilizerness (“magic”) in some settings. We report a worked example, a k -resolved nonstabilizerness map of the $SU(2)_k$ anyon braid-representation family, in which a standard temporal witness is instead systematically *blind*: at $k = 4$ the three-time Leggett–Garg witness saturates its macrorealistic bound exactly ($K_3 = 1.000000$ over every state, braid element, and measurement axis) in the equal-time-step protocol of Table I ($V_1=V_2$), while the single-qubit fusion channel carries near-maximal nonstabilizerness ($M_2 = 0.5585$, 95% of the finite-dimensional ceiling $\log_2(3/2)$). We prove this blindness as a structural theorem: the witness depends only on the Bloch-sphere Gram geometry of the braid orbit, not on nonstabilizerness, and $k = 4$ happens to align the fixed measurement axis with a threefold orbit symmetry (Bloch dot products all $-1/3$) that caps the witness at 1; a finite, Clifford-generating group — the chiral octahedral group, the $k=2$ braid image — with a misaligned axis reaches $K_3 = 3/2$ under a two-propagator protocol. Neither finiteness of the braid image nor the Clifford property is by itself the mechanism. We complement this temporal certificate with three independent certificates of genuine nonstabilizerness — two of them long-range (a doubled-Fibonacci mutual-information witness, $H = 1.700979$, and a gauge-invariant minimum ground-space stabilizer Rényi entropy, ≥ 6.5 against an exact-zero toric-code control), the third a complementary gate-based non-Cliffordness measure — an honest negative control confirming that leakage-free constructions remain strictly additive, and a hardware-oriented single-qubit signal-to-noise prediction. We further pair the $k = 4$ dissociation with Kochen–Specker–Klyachko contextuality at their shared $d = 3$ interface, where the KCBS witness certifies contextuality yet shows only a weak, nongeneric correlation with magic. Two reconciliations with the literature are made explicit: the present result contradicts neither the equivalence of maximal nonlocality and maximal magic established for optimized magic states in a different game, nor the contextuality-supplies-magic theorem; both concern a different object than the fixed, per- k braid generator studied here.

I. INTRODUCTION AND RELATED WORK

Nonstabilizerness, or “magic,” is the resource that lifts Clifford circuits to universal quantum computation [1, 2], and it is by now well established that standard Bell-type correlation witnesses can *detect* it: Macedo *et al.* [3] and Cusumano *et al.* [4] both construct Bell inequalities whose violation certifies nonstabilizerness, and the Ising-anyon no-go of Howard and Vala [5] shows, in the opposite direction, that Clifford-only (Ising) braiding transformations cannot violate a Bell inequality when only topologically protected stabilizer operations are performed. Both directions concern whether, and how, a correlation witness sees magic. A companion paper realizes exactly the Howard–Vala Ising setting microscopically, on a concrete Kitaev honeycomb Hamiltonian, showing that the noncommuting CHSH Bell-operator *algebra* is realized locally while a genuine physical violation is not [27]; the magic resource whose presence or absence that companion paper’s argument turns on is exactly the subject of the present paper, and we cross-reference it bidirectionally throughout.

This paper reports the complementary direction, a *blind spot*: a standard correlation witness — the temporal Leggett–Garg inequality $K_3 \leq 1$ [8, 9] — fails to detect near-maximal single-qubit magic present in a specific, structurally identified member of the $SU(2)_k$ braid-representation family, at $k = 4$. This is not a new claim that correlation witnesses can be blind to magic *in general* — Howard and Vala already establish an instance of witness blindness for Ising braiding [5] — but a k -resolved, numerically exhibited, and structurally explained dissociation map across the $SU(2)_k$ family, together with three independent certificates of genuine nonstabilizerness (two long-range, the third a complementary gate-based measure) and an honest negative control. We stress at the outset what this paper is *not*: it is not a claim that magic is generically decoupled from Bell/CHSH correlations — Macedo and Cusumano show the opposite in their respective constructions — and the present $k = 4$ result is the complementary blind spot on a single, finite braid generator, not a general no-go. A related but distinct construction, Robin and Savage’s anti-flatness witness of nonlocal magic in two-particle scattering [16], cites the same spatial magic–CHSH suppression as Macedo and Cusumano but treats a different Hilbert space (a two-qubit scattering channel, not anyons or an $SU(2)_k$ family) and does not touch the temporal or k -

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resolved question studied here; we note it as related work rather than a competing claim.

An independent line of work develops a general recipe for turning standard tomographic measurements into tailored Bell inequalities and applies it to witness magic: Cieřliński *et al.* [17] show that the same Pauli-basis measurements routinely taken for state tomography certify nonlocality at no additional experimental cost, and, noting the stabilizer structure of the resulting inequalities, use them as magic witnesses. This is again a *detection* result for a spatial (Bell-type) witness, on a general multi-qubit tomographic setting rather than an anyon braid-representation platform; it does not address the temporal Leggett–Garg witness or the k -resolved dissociation that is the subject of the present paper.

Two further reconciliations, closer to the present claim than the detect/blind distinction above, are necessary before we can state the $k = 4$ result without ambiguity. Howard [6] proves an *equivalence* between maximum nonlocality and maximum magic in a qudit CHSH-type game optimized over the magic *state*: this sounds directly opposed to any witness/magic dissociation. The distinction is the object being varied. Howard’s equivalence concerns the optimal single-qudit magic *state* in a fixed correlation game; the present $k = 4$ result concerns a fixed *braid generator* (no state optimization) at one specific value of the family index k , evaluated as a witness of whichever magic that braid generator’s fusion channel happens to carry. The two are different optimization problems over different objects, and neither result constrains the other. Second, Howard, Wallman, Veitch, and Emerson [7] prove that contextuality is necessary for the magic-state resource that promotes stabilizer computation to universality (odd-prime local dimension). We report below (Sec. VIII) a near-zero, nongeneric correlation between a KCBS contextuality witness and magic at $k = 4$; this is not in tension with the HWVE theorem. HWVE is a global existence/threshold statement about universality (odd-prime local dimension), whereas our KCBS correlation is a pointwise measurement over one finite- k orbit, not a threshold claim. The separate blindness of our temporal K_3 witness (Secs. III–IV) is of a different character still: K_3 is a temporal witness, not a spatial/polytope contextuality inequality, and the $k = 4$ dissociation is a structural corollary of a single braid generator being a z -axis rotation, not a general statement about contextuality and magic.

The rest of this paper is organized as follows. Section II fixes five nonstabilizerness axes and the $\log_2$ convention used throughout. Section III presents the k -resolved dissociation map, Fig. 1, with the $k = 4$ blind spot as the central result. Section IV proves the blindness as a structural theorem. Section V reports three independent certificates of nonstabilizerness — two long-range and a complementary gate-based axis. Section VI reports an honest negative control. Section VII gives a single-qubit hardware signal-to-noise prediction. Section VIII maps the dissociation onto an interface with KCBS contextu-

ality, left open here, and states the scope limitations of this work.

II. FIVE NONSTABILIZERNESS AXES

We fix one convention throughout: all magic monotones are reported in base-2 logarithm (bits), following the resource-theoretic construction of Veitch *et al.* [1], $M(\rho) = \log_2(1 + 2\text{sn}(\rho))$, where $\text{sn}(\rho)$ is the sum-negativity of the discrete Wigner representation of ρ . Byles, Forbes, and Pachos [13] report the qutrit gate-magic value $M_2 = \log(16/13)$ without specifying the logarithm base; in base 2 this is $M_2 = \log_2(16/13) = 0.2996$, the value we cite below. The five axes evaluated across the $\text{SU}(2)_k$ family are: (1) the stabilizer 2-Rényi entropy (SRE) M_2 [18], proven to be a genuine magic monotone for Rényi index $\alpha \geq 2$ by Leone and Bittel [19], of the single-qubit braid-generator fusion channel (state magic); (2) the robustness of magic (RoM) [2]; (3) the amortized, gate-based non-Cliffordness of the braid generator itself (Sec. III); (4) the two independent long-range-nonstabilizerness certificates of Sec. V (which additionally re-presents axis (3) as a complementary third non-Cliffordness check); and (5) the temporal Leggett–Garg witness K_3 and the KCBS contextuality witness of Sec. VIII, both used as correlation witnesses rather than magic monotones. An independent, sampling-based route to the same family of monotones is Bell sampling [20], which we do not use directly but note as an alternative, hardware-compatible estimator.

III. THE k -RESOLVED DISSOCIATION MAP

For each k we consider the standard $\text{SU}(2)_k$ braid generator acting on a single fusion-channel qubit as a z -axis rotation by the level-dependent angle $\theta_k = \pi(k+4)/(k+2)$ [10, 11]; the braid image is dense in $\text{SU}(2)$ for every $k \geq 2$ with $k \notin \{2, 4, 8\}$ [12] (density alone, not finiteness, is the relevant classification for the “dense” rows of Table I below). We evaluate, independently, a correlation witness (the temporal Leggett–Garg K_3) and three magic monotones (the stabilizer 2-Rényi entropy M_2 , the robustness of magic RoM, and the amortized gate-based non-Cliffordness of Sec. IV) across $k = 2, \dots, 10$. Table I and Fig. 1 report all four quantities; the three magic monotones, evaluated by an independent re-implementation cross-checked by two further computations sharing verified conventions, give a concordant classification across the family — each sits at its own no-magic value at $k = 2$, lies strictly above it at every higher level, and drops at the two levels whose braid image is finite — although their numerical ranges differ (for M_2 : the Clifford floor 0 at $k = 2$, a finite non-Clifford plateau at $k = 4, 8$, and the dense-representation ceiling $\log_2(3/2) = 0.585$ approached at $k = 3, 5, 6, 7, 9, 10$), and the two K_3 columns (state- and axis-optimized K_3^{opt} , ver-

sus the fixed-measurement-axis value $K_3(Q=\hat{z})$ used operationally in Sec. IV) agree everywhere except at $k=8$, where the two differ by construction (Sec. IV, scope remark).

TABLE I. k -resolved dissociation map. FLW: dense in $SU(2)$ (Freedman–Larsen–Wang universality); for $k \geq 2$, finite braid images occur only at $k \in \{2, 4, 8\}$ [10], corresponding to the binary polyhedral groups $2O, 2T, 2I$ of orders 48, 24, 120 (McKay correspondence with the E_7, E_6, E_8 Dynkin diagrams). K_3^{opt} : Leggett–Garg witness in the equal-time-step protocol ($V_1=V_2$), maximized over states and measurement axis, and over the full braid image for the finite cases $k=2, 4, 8$; for the dense cases by random-word sampling, which approaches but cannot attain the supremum. $K_3(Q=\hat{z})$: same witness at the fixed measurement axis used in Sec. IV. M_2 : stabilizer 2-Rényi entropy (state magic, $\log_2$). RoM: robustness of magic. All values reproduced by an independent re-implementation and cross-checked by two further evaluations sharing verified conventions; $|\Delta| \leq 10^{-6}$ throughout.

k	finite image	K_3^{opt}	$K_3(Q=\hat{z})$	M_2	RoM
2	$2O$ (48)	1.000	1.000	0.000	1.000
3	dense	1.500	1.498	0.585	1.732
4	$2T$ (24)	1.000	1.000	0.5585	1.715
5	dense	1.500	1.479	0.585	1.732
6	dense	1.500	1.495	0.584	1.732
7	dense	1.498	1.494	0.585	1.732
8	$2I$ (120)	1.427	1.342	0.5530	1.714
9	dense	1.500	1.497	0.585	1.732
10	dense	1.500	1.491	0.585	1.732

The central object of this paper is the $k=4$ column of Table I: the Leggett–Garg witness is pinned at its macrorealistic bound, $K_3 = 1.000000$, over *every* state, every braid word, and every measurement axis in the equal-time-step protocol ($V_1=V_2$). The axis-sweep is a genuine numerical search, not a fixed-axis artifact — the same axis-optimization protocol applied at $k=8$ (Sec. IV) finds a strictly larger value there, 1.427 against the fixed-axis 1.342, so the search demonstrably explores nonaligned axes and still finds no $k=4$ violation. The closed-form Gram-geometry argument of Sec. IV independently explains *why* the physical, fixed measurement axis used throughout this paper gives exactly 1 at $k=4$ (it is aligned with a threefold orbit symmetry). In the equal-time-step protocol of Table I ($V_1=V_2$) the $k=4$ blindness holds for *every* measurement axis, a closed result established for the companion $SU(2)_k$ Leggett–Garg witness [28] (its $M(U)$ eigenvalues exceed 1 only for rotation angles in $(0^\circ, 90^\circ)$, which the tetrahedral group does not contain); under the more general two-propagator protocol of Sec. IV the blindness is instead bound to alignment, and a misaligned axis reaches $K_3 = 3/2$ — the alignment mechanism itself, not a defect.

Meanwhile, the maximum state magic accessible over the $k=4$ braid-image orbit reaches $M_2 = 0.5585$, i.e. 95% of the finite-dimensional ceiling $\log_2(3/2) = 0.585$ approached by the dense representations, and a robust-

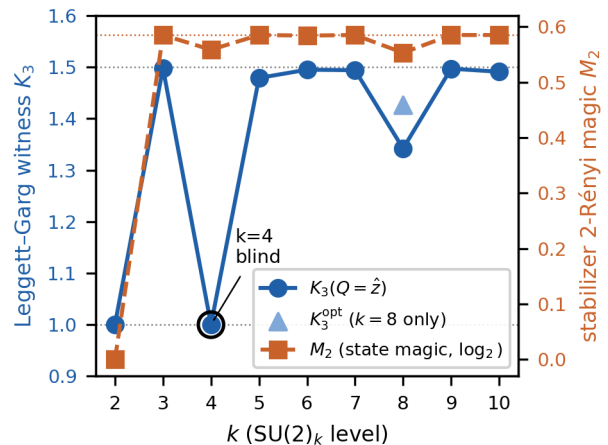


FIG. 1. The $k=4$ dissociation. Top: Leggett–Garg witness $K_3(Q=\hat{z})$ (blue) against stabilizer 2-Rényi magic M_2 (orange, $\log_2$ basis) across $k=2, \dots, 10$. At $k=4$ the witness sits at its macrorealistic ceiling, $K_3 = 1.000$ (open circle, blind), while $M_2 = 0.5585$ is at 95% of the dense-representation ceiling $\log_2(3/2) = 0.585$ (dashed line). At $k=2$ the witness is likewise at its bound, but there is no magic to miss ($M_2 = 0$); at every dense k and at $k=8$ it exceeds the classical bound. The dissociation is specific to $k=4$: witness silent, resource present. The $k=8$ point shows the fixed-axis value $K_3(Q=\hat{z})$; the axis-optimized value $K_3^{\text{opt}} = 1.427$ (Table I) is shown as a lighter marker for comparison.

ness of magic RoM = 1.715 close to the dense ceiling $\sqrt{3} = 1.732$. We stress the framing: $K_3 = 1$ at $k=4$ is a *single-qubit corollary* of a specific braid rotation angle, not a new physical phenomenon, and the finding is that the witness misses a resource that is, by every available magic monotone, almost as large as it can be for this family.

IV. STRUCTURAL THEOREM: ALIGNMENT, NOT FINITENESS OR CLIFFORD

We now explain the $k=4$ blind spot as a theorem rather than a numerical coincidence. Consider the temporal Leggett–Garg protocol with a fixed dichotomic observable $Q = \hat{n}_0 \cdot \sigma$ (here $\hat{n}_0 = \hat{z}$) and two independent propagators V_1, V_2 drawn from a group $G \subset SU(2)$, giving Heisenberg observables $Q_1 = Q$, $Q_2 = V_1^\dagger Q V_1$, $Q_3 = (V_2 V_1)^\dagger Q (V_2 V_1)$ with Bloch directions $\hat{n}_1 = \hat{n}_0, \hat{n}_2, \hat{n}_3$ in the orbit of $\hat{n}_0$ under the $SO(3)$ image of G . For any single-qubit dichotomic pair, $\{\hat{n} \cdot \sigma, \hat{m} \cdot \sigma\} = 2(\hat{n} \cdot \hat{m})\mathbb{1}$, so the Lüders two-time correlator reduces to a Bloch dot product independent of the state,

$$C_{ij} = \hat{n}_i \cdot \hat{n}_j \quad \text{for every } \rho, \quad (1)$$

and consequently $K_3^{\text{max}} = \max_{(\hat{n}_2, \hat{n}_3) \in \text{orbit}} (\hat{n}_1 \cdot \hat{n}_2 + \hat{n}_2 \cdot \hat{n}_3 - \hat{n}_1 \cdot \hat{n}_3)$ depends only on the Gram matrix of Bloch

dot products within the orbit of $\hat{n}_0$ under G — *not* on whether the braid representation is finite, and not on whether G is a Clifford subgroup. The real $k = 4$ braid image is the binary tetrahedral group $2T$ (projective order 12), and σ_1 acts as a projective threefold rotation about $\hat{z}$: the orbit of $\hat{z}$ under the full projective group $2T$ is exactly the four vertices of a tetrahedron, with every off-diagonal Bloch dot product equal to $-1/3$. Exhaustive enumeration over the $4^2 = 16$ orbit pairs (with $\hat{n}_1 = \hat{n}_0$ fixed) gives $K_3^{\max} = 1$ exactly, matching Table I. The same Gram argument applied to the real $k = 8$ braid image (binary icosahedral group $2I$, twelve icosahedral Bloch directions, dots $\{\pm 1, \pm 1/\sqrt{5}\}$) gives $K_3^{\max} = 3/\sqrt{5} = 1.341641$ in the fixed- $\hat{z}$ -axis protocol, and $K_3^{\text{opt}} = 2 \cos 72^\circ - \cos 144^\circ = 1.427$ under the axis-optimized protocol in which the measurement axis is rotated to be perpendicular to a fivefold orbit axis unreachable from the fixed $\hat{z}$ frame — the two $k = 8$ numbers in Table I differ by measurement-protocol scope, not by inconsistency.

The corollary that gives this theorem its bite is a direct counterexample to the plausible-sounding slogan “a finite braid image or a Clifford-generating group is witness-blind”: the chiral octahedral group (24 elements, itself a genuine single-qubit Clifford subgroup) with the measurement axis *rotated* to $\hat{n}_0 = (1, 1, 0)/\sqrt{2}$ instead of a coordinate axis has orbit dots $\{0, \pm \frac{1}{2}, \pm 1\}$ and reaches $K_3^{\max} = 3/2$ exactly, the full Lüders bound — realized operationally by two genuine $\text{SU}(2)$ qubit unitaries reproducing the same value to machine precision. The same finite, Clifford-generating group is witness-blind at one axis and witness-saturating at another; neither finiteness nor the Clifford property is the driver. The same conclusion follows across levels rather than across axes (Fig. 2): the three levels with a finite braid image do not share a verdict, their group orders are not ordered along k , and the two levels closest in magic lie on opposite sides of the bound. What is sufficient for blindness is *alignment*: the fixed measurement axis coincides with a symmetry axis of the orbit whose Gram structure caps the witness (here, $\hat{z}$ is the threefold axis of the $k = 4$ tetrahedral orbit, with all Bloch dot products $-1/3$), forcing the orbit Gram matrix into a form whose maximum is exactly 1. We emphasize, following the explicit abstraction argument, that Gottesman–Knill classical simulability of Clifford circuits is not invoked anywhere in this argument and does not by itself imply Leggett–Garg inertness; the mechanism is the Bloch-sphere Gram geometry of the orbit, evaluated in Eq. (1), not computational-complexity simulability. Correspondingly, the correct statement of the $k = 4$ result is that *this witness measures orbit alignment, not magic or universality*; it says nothing about other witnesses or other measurement axes.

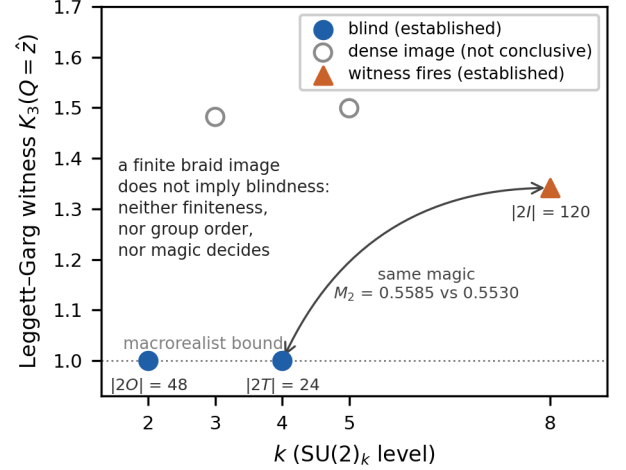


FIG. 2. What does *not* decide blindness. Fixed-axis Leggett–Garg witness $K_3(Q=\hat{z})$ against the level k , for the three levels with a finite braid image ($k = 2, 4, 8$; filled markers, group orders in the binary convention of Table I) together with two dense levels shown for reference ($k = 3, 5$; open markers, sampling-dependent and therefore not conclusive). The orders are not ordered along k ($48 \rightarrow 24 \rightarrow 120$), and $k = 4$ and $k = 8$ carry nearly the same magic ($M_2 = 0.5585$ against 0.5530) while lying on opposite sides of the macrorealistic bound. The dense levels are kept in the figure because they are the evidence for their own exclusion: among the three finite levels alone, other columns of the deposited data would appear to separate the verdicts. What does decide — alignment of the fixed axis with an orbit symmetry axis whose Gram structure caps the witness — is stated in the text above; no column of the deposited data carries it as a scalar, so it is not drawn here. All plotted values are read from p5a_ergebnis_t4.json.

V. THREE CERTIFICATES OF GENUINE NONSTABILIZERNESS

Beyond the single-qubit dissociation of Secs. III–IV, we report three methodologically independent certificates of genuine nonstabilizerlessness, in the same spirit as the exact $\text{SU}(2)_k$ mana computation of Fliss [14], the exact-setting anchor we follow most closely among prior anyon-magic constructions. The first two, (1) and (2), certify genuine *long-range* (many-body, topologically encoded) magic not reducible to a local artifact; the third, (3), is the complementary gate-based non-Cliffordness axis of the map (axis (3), Sec. II) — a single-generator quantity included here as a methodologically distinct third line of non-Cliffordness evidence, not a long-range-state certificate in the sense of (1)–(2). Each of the two long-range certificates is a *sufficient* condition for genuine long-range magic (a noninteger witness value, or a gauge-invariant lower bound bounded away from zero); we make no claim that either is *necessary*, and do not rule out other long-range-magic witnesses that these two constructions hap-

pen to miss.

These certificates are not offered as a priority claim on the presence of long-range nonstabilizerness in this class — for the two long-range certificates (1) and (2) that statement is the theorem of Zhang *et al.* [22] quoted below — but as the quantitative substrate for the dissociation of Secs. III–IV: they establish, with explicit values and null controls, the magic that the temporal witness demonstrably fails to detect. Certificate (3) is not covered by that theorem at all: it is a gate-based measure, whereas Zhang *et al.* characterize ground states — and it is the axis most directly tied to the blind spot, since the same generator that leaves $K_3 = 1.000$ carries gate magic 0.765.

(1) *Doubled-Fibonacci mutual-information witness.* Following the long-range-nonstabilizerness mutual-information construction of Korbany *et al.* [21], we evaluate the Shannon entropy of the topological-sector weight distribution on the doubled-Fibonacci ground space, $d_{(a,b)} = d_a d_b$ for $(a,b) \in \{1,\tau\} \times \{1,\tau\}$: with quantum dimensions $d = (1, \varphi, \varphi, \varphi^2)$, $\varphi = (1 + \sqrt{5})/2$, the normalized weights are $p = d^2/D^2$ with $D^2 = \sum_a d_a^2 = 1 + 2\varphi^2 + \varphi^4 \approx 13.09$, giving

$$H = - \sum_a p_a \log_2 p_a = 1.700979, \quad (2)$$

a noninteger value that certifies genuine long-range magic (Korbany’s Theorem 3: an integer value would indicate a stabilizer ground space and no certificate). We obtain the same value from a second, independently sourced input: the vacuum column of the doubled-Fibonacci modular S -matrix extracted from a 3×3 lattice (Wilson-loop plus C_3 -rotation channel) [30], with no analytic quantum dimensions and no F/R symbols entering the extraction. The normalized weights $|S_{0a}|^2$ from that column give the same quantity; the two evaluations agree to $|\Delta| = 1.8 \times 10^{-11}$, at the level of the lattice-extraction residual. What this establishes is provenance: both routes evaluate the same witness functional, and their agreement shows that its input can be reconstructed from lattice data alone rather than imported from the analytic category data. (The closed-form rearrangement $H = \log_2 D^2 - \sum_a p_a \log_2 d_a^2$ is algebraically identical to the direct sum and is retained only as an internal consistency check, not as an independent route.) The extracted vacuum column and its evaluation script are included in the deposit. As a null control on the same witness construction, the $\mathbb{Z}_2$ toric-code ground space (four sectors of equal quantum dimension $d = 1$) gives the integer value $H = 2.000000$ exactly, confirming the witness correctly reports *no* certificate on a stabilizer code; a second null anchor, the binary-entropy toric-code T -state of Korbany’s Eq. (3.30), $H_2(\cos^2 \frac{\pi}{8}, \sin^2 \frac{\pi}{8}) = 0.600876$, is also reproduced to the digits shown (an earlier candidate value 0.5436 for this same anchor is confirmed incorrect).

Remark (scope). We evaluate the witness analytically on modular data extracted from the lattice as described

above. The statement applicable to our model is Theorem 3 of Ref. [21], formulated directly for string-net ground states [their Eq. (4.2)]: it certifies long-range nonstabilizerness whenever $H(\{|\psi_a^\zeta|^2\}) \notin \mathbb{N}$ for some modular frame ζ , with M_ζ an arbitrary product of S and T . Our evaluation is the case $\zeta = S$ applied to the vacuum minimal-entropy state, giving $H = 1.700979$. Theorem 3 imposes no conditions on the regions; the authors note only that for a given ζ the system must be large compared with the depth of the corresponding modular circuit, and observe that this is not a restriction because long-range nonstabilizerness is an asymptotic property of a family of states. (The quantitative width and separation conditions are given in Lemma 2 of that reference, which is stated not for a specific model but for any state whose reduced density matrix factorizes as in their Eq. (4.5) — the form derived there for string nets.) The region condition stated for string nets is that A_γ and B_γ be disjoint and “sufficiently large to support the lattice WLO $W_a(\gamma)$ ”, whose support — unlike in the toric code — extends beyond γ onto neighboring edges and is not quantified there. Independently of these conditions, the geometry itself requires four transverse bands on the torus — two annuli and two non-empty gaps — hence $L \geq 4$ ($E = 48$), or $L \geq 6$ ($E = 108$) if the Wilson-loop ribbon occupies two plaquette rows. At the string-net fixed point ($\xi = 0$) we note that the factorization hypothesis of their Lemma 2 holds exactly, rather than asymptotically, once the regions are separated beyond the support of the local terms. The lattices used here ($E = 12$, $E = 27$) do not host even the minimal band structure. A direct evaluation of the mutual information is therefore not obstructed in principle, but the smallest admissible size lies a factor $\geq 2^{21}$ in Hilbert-space dimension beyond exact diagonalization; it is tensor-network work, and we do not attempt it here.

(2) *Gauge-invariant minimum ground-space stabilizer Rényi entropy.* A representative-dependent estimate of the stabilizer 2-Rényi entropy on the doubled-Fibonacci ground space gives $M_2 = 6.393$ (Monte Carlo over 1.5×10^5 Pauli samples, standard error 0.13, in the same reduced-Pauli-string-sampling spirit as the many-body SRE estimator of Ding, Wang, and Yan [25]) — a Monte-Carlo estimate, not a rigorous bound. Scanning instead over 11 representatives of the four-dimensional ground space (gauge-invariant construction; condensate, random, and eigenbasis types, Monte-Carlo standard errors 0.05–0.53) gives a range $M_2 \in [6.5, 8.9]$ across the sampled representatives, all bounded away from zero by more than an order of magnitude. We do not claim a global minimum over the continuous ground space. As a discriminator, the fraction of Pauli-expectation samples landing exactly in $\{0, \pm 1\}$ (the stabilizer signature) is 100% for the $\mathbb{Z}_2$ toric-code control (where the corresponding $M_2 = 0$ exactly) and only 54% for the doubled-Fibonacci ground space, confirming the qualitative long-range-nonstabilizerness diagnostic advocated by Wei and Liu [23] in a related long-range-magic complexity frame-

work, evaluated here on a distinct family (doubled-Fibonacci vs. their surveyed codes and phases); we do not import any specific numeric value from Ref. [23].

(3) *Complementary gate-based non-Cliffordness.* As the gate-based axis of the map (axis (3), Sec. II), distinct from the state magic of (1)–(2), we evaluate the amortized, gate-based stabilizer Rényi entropy of the braid generator itself (rather than of a fixed state), $M^A(\sigma_2)$, across the family: 0, 0.562, 0.765, 0.832, 0.815 for $k = 2, 3, 4, 5, 8$ respectively, with the closed-form single-qubit anchor $M_2^A(T\text{-gate}) = 2 - \log_2 3 = 0.4150$ (agreement to 3×10^{-15} against three independently constructed anchors: the product state $|0\rangle$, the Bell state, and the tensor product $T|+\rangle \otimes T|+\rangle$, the last matching $2\log_2(4/3)$ exactly). This gate-based measure is nonzero already at $k = 3$ and grows through $k = 5$ before slightly receding at $k = 8$, structurally distinct from the state-magic plateau of Table I and consistent with it throughout (both vanish only at the Clifford point $k = 2$).

The general statement is due to Zhang *et al.* [22], who prove — for states related by a constant-depth local circuit to a ground-state wavefunction satisfying the entanglement-bootstrap axioms, of which string-net ground states are the canonical case — that stabilizer states, even up to constant-depth local unitaries, cannot realize the ground states of non-Abelian string-net models. What the present section adds is not that existence statement but its explicit evaluation on a concrete model: the values reported above, together with the $\mathbb{Z}_2$ toric-code null controls.

VI. HONEST NEGATIVE CONTROL: LEAKAGE-FREE ADDITIVITY

As a negative control accompanying the three certificates above, we verify that a leakage-free construction of the same family remains strictly additive under composition: the construction procedure itself produces no genuine long-range or entangling magic beyond what is already present locally (entanglement contribution $n_{\text{ent}} = 0$; product-state consistency error $\sim 10^{-15}$). On the same construction, the single-qubit Wigner-Mana sanity values across $k = 3, 4, 5, 8$ (base 2, $d = 5$ discrete Wigner representation) are 1.0623, 1.0328, 1.0746, 0.9708, all consistent with the plateau structure of Table I and none suggesting an anomalous long-range contribution beyond the additive baseline; the $k = 2$ construction falls outside the $d = 5$ representation used here and is omitted from this specific check. We report this explicitly as a negative result: it rules out one plausible mechanism for spurious long-range magic (leakage across the composition procedure), it does not itself certify anything positive, and the positive certificates of Sec. V stand independently of it.

VII. HARDWARE-ORIENTED SINGLE-QUBIT BENCHMARK

As a single-qubit anchor for a possible hardware signal-to-noise estimate, the closed-form T -gate value $M_2^A(T) = 2 - \log_2 3 = 0.4150$ (Sec. V, item 3) provides a magic-monotone scale against which a device's discrimination of a magic state from the nearest stabilizer state could be benchmarked. We state this explicitly as a *prediction*, not a hardware result: on real hardware the magic state is not distinguishable from a pure T -gate output by this monotone alone (both saturate the same single-qubit value), and no multi-qubit or topological hardware claim is made here; extending this anchor to a genuine multi-qubit hardware demonstration is a distinct, separately resourced undertaking outside the scope of this paper. Non-Abelian Fibonacci-anyon braiding has already been demonstrated on a superconducting processor [15], but that demonstration addresses universality, not a magic-monotone measurement; closing this gap on real hardware remains open. Universal topological gates have now been demonstrated on qubit hardware for the S_3 quantum double, with universality supplied by fusion as a primitive and a topologically prepared magic state [29] — consistent with the magic-not-braiding resource split mapped here.

VIII. INTERFACE WITH CONTEXTUALITY AT $d = 3$, AND LIMITATIONS

We map the $k = 4$ dissociation onto an interface with Kochen–Specker–Klyachko (KCBS) contextuality [24, 26] at qutrit dimension $d = 3$, evaluated on a fusion-channel qutrit introduced in this section. At $k = 4$ the KCBS witness reaches 2.0595, exceeding the classical bound 2 and certifying contextuality (in contrast to the Leggett–Garg witness of Sec. III, which is blind at the same k); at $k = 3, 5, 8$ the witness reaches 2.22, 2.23, 2.23, approaching the quantum bound $\sqrt{5} = 2.2361$. We make no priority claim on braiding exhibiting KCBS contextuality as such: Chou [24] already reports state-dependent contextuality violations for Fibonacci and $SU(2)_k$ braid representations; the contribution here is the pairing of that witness with the magic monotones of this paper at the same k . Evaluated against the qutrit Wigner-Mana value across the orbit at each of $k = 3, 4, 5, 8$, the Pearson correlation between the KCBS witness and magic is weak and nongeneric ($-0.03, -0.19, -0.07, -0.01$ respectively; a weak negative pointwise association at every k), which we read as evidence against a generic one-to-one correspondence between this contextuality witness and magic on this family, evaluated pointwise over a single finite k — not as a claim that no such correspondence could hold more generally, and not in tension with the HWVE theorem discussed in Sec. I for the three reasons given there. The $d \geq 3$ question of whether *any* correlation witness can be proven to always detect magic on this family is

left to future work; the present paper contributes only the magic side of that interface.

We close with the limitations of the present scope. First, we do not claim a two-dimensional area-law scaling of long-range magic; established methods do not currently reach system sizes beyond roughly 27 qubits by exact or efficient sampling methods, so no such scaling claim is made here. Second, the hardware prediction of Sec. VII is a single-qubit anchor only; no multi-qubit hardware claim is made. Third, we do not claim a new genuinely temporal magic witness: the naive temporal Leggett–Garg witness studied here is demonstrably blind at $k = 4$ (Secs. III–IV), and constructing a temporal witness that is not subject to the same alignment blind spot is an open problem we do not attempt to solve. Fourth, a doubled-Ising extension of the family studied here (nominal $M_2 \approx 5.3$ under a $\log_3$ convention) is deliberately excluded from the present, deliberately minimal paper; we checked the two most relevant candidate anchors in the literature, the non-Abelian long-range-magic no-go theorem of Zhang *et al.* [22] and the long-range-nonstabilizerness complexity framework of Wei and Liu [23], and found neither reports an Ising, $\log_3$, or comparable value, so no prior anchor currently supports including it as a load-bearing result, and we defer it to future work.

DATA AND CODE AVAILABILITY

The reproduction scripts and raw JSON outputs supporting the values reported in this paper are deposited at [doi:10.5281/zenodo.21134454](https://doi.org/10.5281/zenodo.21134454). Paper, figures, and data are released under the Creative Commons Attribution 4.0 International (CC BY 4.0) license; the deposited source code is released under the Apache License, Version 2.0 (see LICENSE and LICENSE-CODE). The deposited outputs include the gauge-invariant ground-space stabilizer-Rényi-entropy range $M_2 \in [6.5, 8.9]$ of Sec. V(2) (`p5a_tsre_robust_m2_range.json`, with its provenance block); the underlying exact-diagonalization construction of the ground space is a separate, heavier computation and is not part of this record. The deposited outputs are of two kinds: outputs that a deposited script regenerates (`p5a_ergebnis_t3/t4/t5.json` and `p5a_ergebnis_hardening_b.json`, written by the corresponding reproduction scripts), and value-and-provenance records whose generating computation is heavier than the record itself (`p5a_tsre_robust_m2_range.json` and `p5a_korbany_lattice_vacuum.json`, each with its provenance block). The mutual-information witness value of Eq. (2) is recomputable directly from the

quantum dimensions given there.

USE OF AI TOOLS

In the research, processing, and writing of this paper and its results I worked together with generative AI tools, in practice a system of multiple coordinated AI instances that I set up and orchestrate (large language models, mainly Claude, by Anthropic, inside Claude Code). At their current context sizes I found it far more effective to work with several specialized instances, each with its own role and its own harness of rules and parameters that I designed and refined through feedback, than to load a single instance with all of the material; for my workflow that would have been inefficient, though this depends on the individual implementation. I lead this collaboration: I choose the research directions, set the goals, and make the final decisions in open exchange with the AI, learning actively as the work proceeds. The AI carries out the drafting, including the mathematical and technical parts, the numerical computation, and the literature search, under my direction. The AI works autonomously only task by task, within the structure I develop through feedback: it completes a task, and at open questions that need me it stops until the point is settled before the next step. Along the way I witness and take many of the decisions that shape the path, and it is common for me to spot things that need improvement. The work spans many separate runs, and a single simulation or build task alone can take up to an hour, so it could not happen all together in one autonomous run; and had I let the AI do all of it together alone, even if it is possible, it would no longer be my work but the AI's.

I run multiple verifications at the different stages of the work and one before release, including cross-checks with an unrelated AI model from a different company, and all references are checked against the original sources. In the end what matters are human eyes, a principle that is itself written into the parameters of my system: I reach out to experts after publishing for review and feedback, so I learn what is solid and what must be corrected or falsified. My scripts for reproduction and review are released with this record. These tools are not authors; I am the author, and I take full responsibility for all scientific content and decisions leading to these results and their publication.

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